1. Hardware schematic and documentation

Requested an AI-generated schematic showing actual Raspberry Pi 4, 4N25 optocoupler, SN754410NE H-bridge, motor, supplies, resistors, and capacitor connections, plus assembly instructions and a parts list.

2. Pre-power safety review

Requested guidance for inspecting power rails, shared grounds, H-bridge pinout, fail-safe motor enable, and safe initial power-on. Student-written inspection must reflect actual measurements.

3. Motor control software

Requested help developing/debugging Raspberry Pi C and ARM assembly routines for motor enable, direction, stepping, timing, and keyboard controls required by CP1.

4. Electrical verification

Discussed GPIO logic voltages, optocoupler enable behavior, motor winding identification and resistance, and stepping patterns for clockwise/counterclockwise commands.

5. Circuit corrections

Clarified that four 1.8 kΩ resistors were the driver-input pulldowns, that bypass capacitors were installed, and that direction reporting should distinguish electrical pattern verification from physical motor confirmation.

6. Motor performance and demonstration

Discussed one-revolution and continuous-rotation demonstrations, motor stand, stall-speed measurements and reporting, including three trials and the 200 RPM threshold.

7. Android app and Bluetooth roadmap

Chose MIT App Inventor for Android and requested a checkpoint-by-checkpoint work plan for Bluetooth messaging, motor controls, IR RPM, AS5600 encoder, calibration, logging, Maintain mode, and GitHub documentation.